LARA BENCHMARK · LOCAL EVALUATION STUDY

Can we trust a small local LLM-as-judge?

A 2×2 **generator × judge** study — measuring how much the grader, rather than the model under test, drives the accuracy number.

RAG mode | book · 32k context | N = 10 / task × 4 tasks | qwen2.5 — 7B vs 14B | local · Apple M2 · 16 GB

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THE PROBLEM

Generation works.
Scoring doesn't.

The LaRA paper grades answers with a very large judge (GPT-4o / Qwen-Max). A weak **local** judge tends to answer “False” even when the answer is correct — so every accuracy number collapses to **0.0**. The pipeline is right end-to-end; the grader is the bottleneck.

> **So how much does the judge's size actually change the verdict?**

METHOD — A 2×2

Vary two things independently

	Judge 7B	Judge 14B
Gen 7B	CELL 1 weak / weak	CELL 2 weak / strong
Gen 14B	CELL 3 strong / weak	CELL 4 strong / strong

Answers are generated **once per generator**, then graded by **both** judges — so the two judges see identical answers. That isolates the judge's effect. Everything else is fixed: RAG, book, 32k, 10 questions × 4 task types (location, reasoning, comparison, hallucination).

THE HEADLINE

52.5% → 70.0% +17.5pts

The **identical** 14B answers, graded by a 7B judge vs a 14B judge. Nothing about the model under test changed — **only the grader did.**

RESULT 1 — ACCURACY

The judge moves the score more than the model

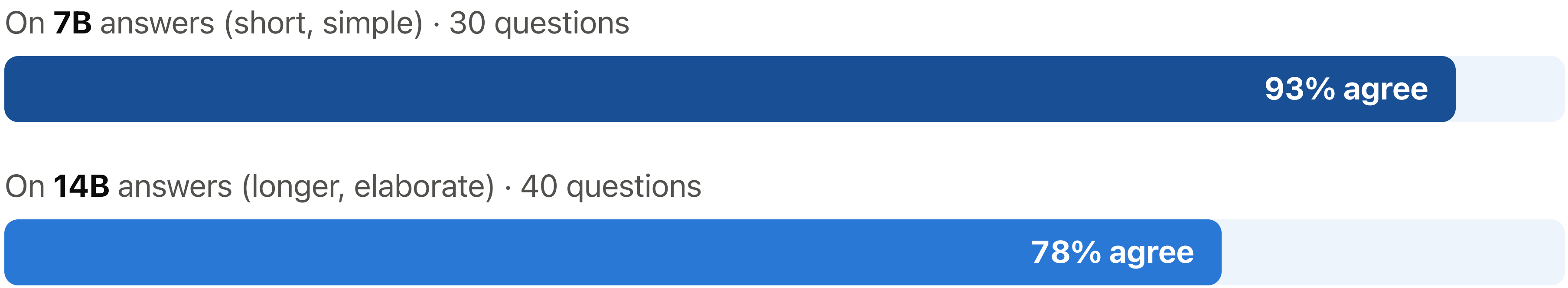
	Judge 7B	Judge 14B
Gen 7B	66.7 3 tasks	66.7 3 tasks
Gen 14B	52.5 weak judge	70.0 strong judge

GEN / JUDGE	LOC	REAS	COMP	HALLU	OVERALL
7B / 7B	70	40	n/a	90	66.7
7B / 14B	60	50	n/a	90	66.7
14B / 7B	60	40	10	100	52.5
14B / 14B	60	**80**	**50**	90	**70.0**

On the 3 shared tasks, the strong judge separates the models (**14B 76.7 > 7B 66.7**) — but the weak judge calls them a **tie (66.7 each)**. A weak judge can erase the ranking the benchmark exists to measure.

RESULT 2 — JUDGE AGREEMENT

The weak judge is too strict — worse on longer answers



8

the 7B judge was **too strict**
(failed an answer the 14B judge passed)

1

too lenient
(the reverse)

WHAT IT MEANS

Three takeaways

1. The **judge moves the score more than the generator does** — +17.5 points on identical answers, just by swapping the grader.
2. The weak judge is **asymmetrically too strict** (8 vs 1), and its unreliability **grows with answer length** (93% → 78% agreement).
3. Generation can stay local, but **grade with a strong (API) judge**. Local accuracy with a small judge is a **lower bound**, not a true score.

NOTES

Infrastructure & limitations

Making the run fit in 16 GB

- The 2×2 study first **OOM-killed** the machine: the RAG eval loaded a fresh embedding model + reranker **per question** across 8 workers (~1.4 GB each) on top of Ollama's 9 GB 14B model.
- Fix: a process-wide **model cache** loads each retrieval model once; concurrency lowered to 2.
- Peak dropped from **20 GB+ to ~10 GB** — the run completes (on swap during the 14B phase).

Limitations

- Single slice: RAG / book / 32k only. No Long-Context comparison; **128k does not fit in 16 GB** (needs ASU compute).
- Both judges are **local and small** — agreement bounds consistency, not ground-truth correctness.
- **N = 10** per task type — directional, not statistically tight.
- **7B / comp** cell is n/a (a transient memory failure during the run).